J Cancer Res Clin Oncol (2013) 139:557–562
DOI 10.1007/s00432-012-1357-1

ORIGINAL PAPER

Midkine mRNA level in peripheral blood mononuclear cells is a novel biomarker for primary non-small cell lung cancer: a prospective study

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Received: 28 August 2012 / Accepted: 23 November 2012 / Published online: 7 December 2012

Abstract

Background and objectives Midkine (MK) mRNA was highly expressed in various human cancer tissues and cells. The present study aimed to investigate whether MK mRNA level in peripheral blood mononuclear cells (PBMC) could serve as a diagnostic biomarker for patients having primary non-small cell lung cancer (NSCLC).

Methods MK mRNA level in PBMC from 87 patients with primary NSCLC, 35 patients with lung benign lesion (LEL), and 30 healthy volunteers was analyzed by real-time quantitative RT-PCR. The levels of serum carcino-embryonic antigen, carbohydrate antigen 125 (CA125), and neuron-specific enolase were detected by chemiluminescent microparticle enzyme immunoassay.

Results PBMC MK mRNA level was significantly higher in patients with primary NSCLC than that in other groups ($P < 0.001$), while there was no significant difference between LEL patients and healthy volunteers ($P > 0.05$). Higher MK mRNA level was correlated with clinical stages ($P = 0.026$), differentiation ($P = 0.025$), and lymph node metastasis ($P = 0.022$) of NSCLC. Using a cutoff of 0.0063, the sensitivity and specificity of MK mRNA levels to differentiate between patients with NSCLC and patients with LEL were 57.47 and 93.33 %,and it were 56.32 and 93.33 % for patients with NSCLC and healthy volunteers, respectively. Furthermore, multivariate analysis indicated that PBMC MK mRNA level above the cutoff value presented a chance of 11-fold higher for NSCLC occurrence.

Conclusions MK mRNA level in PBMC may be a potential non-invasive molecular marker for the diagnosis of primary NSCLC.

Keywords Midkine · Diagnosis · Non-small cell lung cancer · Tumor biomarkers · PBMC

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Introduction

Lung cancer, with the highest incidence rate and death rate, is the most common malignant tumor in the world. Despite the improvements in the diagnosis and treatment in recent years, the 5-year survival rate of lung cancer remains as low as 16 % (Jemal et al. 2010). The most widely used lung cancer biomarkers are carcinoembryonic antigen (CEA), cancer antigen 125 (CA 125), cyfra21-1(cytokeratine 19 fragment), and neuron-specific enolase (NSE) (Molina et al. 2008). However, to date, no validated tumor marker is sufficiently accurate to be useful in diagnostic tests. Therefore, searching for novel biomarkers remains a tough topic in NSCLC diagnosis.

Midkine (MK) is a member of the heparin-binding growth factor family, which was originally identified as the product of a retinoic acid-responsive gene during embryogenesis. MK is strongly expressed during the mid-gestation but its expression is highly restricted in normal tissues of adults (Kadomatsu et al. 1988; Tomomura et al. 1990). However, recent studies have shown that MK mRNA is highly expressed in various human cancer tissues and cells and proposed as a potential biomarker of malignant tumors (Aridome et al. 1995; Garver et al. 1993; Ohhashi et al. 2009). Ohhashi et al. (2009) showed that high MK mRNA may be useful as a new diagnostic and prognostic biomarker in predicting malignant properties of pancreatic cancer. In addition, increased MK expression in peripheral blood offered particular promise as a prognostic marker in children with B-precursor acute lymphoblastic leukemia (Hidaka et al. 2007). However, it remains unknown whether MK mRNA level in peripheral blood could serve as an effective biomarker for NSCLC patients.

In this report, a sensitive and specific quantitative Taqman real-time PCR was utilized to determine the MK mRNA level in peripheral blood mononuclear cells *(PBMC)* from patients with NSCLC, patients with LEL, and healthy controls. Furthermore, the relationships between MK mRNA level and the clinical characteristics of NSCLC patients were evaluated.

Patients and methods

Patients

This is a prospective study that was approved by the Ethics Committee of Huzhou Centre Hospital, and all participants gave informed consent. A total of 87 patients with primary NSCLC who were consecutive patients admitted to Huzhou Centre Hospital from March 2009 to May 2011 were enrolled in the study, and their characteristics was shown in Table 1. All patients were diagnosed without, history of other epithelial tumors or infectious diseases. The diagnosis of NSCLC was based on iconography and pathology. Tumor stage and grading were classified according to the tumor node metastasis (TNM) staging system (Rami-Porta et al. 2009). As controls, 35 patients with LEL including bronchiectasis disease (eight cases), pulmonary tuberculosis (five cases), pulmonary inflammatory pseudotumor (seven cases), and chronic obstructive pulmonary disease (15 cases), and 30 healthy volunteers were enrolled.

Table 1 The correlation of MK mRNA level in PBMC with clinicopathological features of NSCLC patients

Parameter	n	Median MK mRNA in PBMC $2^{-\Delta CT}$/(E−03)	Z	P value
Sex				
Male	55	7.27 (3.91–11.41)	−0.009	0.993
Female	32	6.20 (3.14–15.62)		
Age				
<60	31	7.20 (3.23–15.99)	−0.363	0.716
>=60	56	7.12 (3.84–11.22)		
Smoking				
Yes	48	9.79 (5.20–14.24)	−1.613	0.107
No	39	4.49 (3.51–8.39)		
Histology[a]				
Adenocarcinoma	35	7.74 (4.15–15.99)	−1.651	0.099
Squamous cell	48	6.54 (3.29–9.89)		
TNM stage				
I–II	23	4.99 (3.51–7.74)	−2.233	0.026
III–IV	64	8.52 (4.00–17.13)		
Differentiation				
Well-moderate	53	5.92 (3.77–9.36)	−2.236	0.025
Poor	34	10.53 (4.72–22.69)		
Tumor size				
T<=3 cm	18	8.16 (2.82–24.19)	−0.954	0.340
$T > 3$ cm	69	10.02 (2.61–28.17)		
Lymph nodes metastasis				
No	65	6.32 (3.86–10.03)	−2.295	0.022
Yes	22	16.35 (3.51–24.20)		

[a] Four patients with large cell carcinoma were not included in this table

Collection of PBMC

A total of 10 ml of peripheral blood in EDTA was obtained in the middle of vein puncture in order to avoid contamination of blood with skin epithelial cells. PBMCs were separated by gradient density centrifugation using Ficoll-Hypaque 1077 (Sigma) at 400×g for 30 min at room temperature, then interface cells were removed and washed twice with 10 ml PBS (pH 7.2). PBMCs were stored at −80 °C until the use for RNA extraction. The whole procedure for the collection of PBMC was processed within 2 h.

Real-time quantitative PCR

Total RNA was isolated from PBMC using Trizol LS reagent (Invitrogen) according to the manufacturer's instructions. RNA concentration was quantified by SmartSpec™ Plus (Bio-Rad), and RNA integrity was checked with electrophoresis by observing the 28S and 18S RNA bands. Reverse transcription of RNA was carried out with the PrimeScript™ RT Kit (TaKaRa), and cDNA was stored at −80 °C for further analyses.

To avoid genomic DNA contamination, probes were designed to span exon–exon junctions using Primer Express software version 2.0 (Applied Biosystems, USA). The primers and probes were synthesized by GeneCore BioTechnologies (Shanghai, China) with the following sequences: for Midkine Forward primer: 5′-GACCATCCG CGTCACCA-3′, Reverse primer: 5′-TCCAGGCTTGGC GTCTAGTC-3′, Taqman probe: 5′- FAM-CAAAGGCCA AAGCCAAGAAAGGGAAG-TAMRA-3′; for endogenous control glyceraldehyde-3-phosphate dehydrogenase (GAPDH) (Liu et al. 2005) Forward primer: 5′ –GCCAGCCGA GCCACAT-3′ Reverse primer: 5′- CTTTACCAGAGT TAAAAGCAGCCC-3′, Taqman probe:5′-FAM-CCAAA TCCGTTGACTCCGACCTTCA-TAMRA-3′. PCR was performed with Premix Ex Taq™ kit (TaKaRa) in a final reaction mixture of 20 μl containing 1 μl cDNA, 10 μl 2 × Premix Ex Taq™, 0.4 μl each 10 μM forward and reverse primer, 0.8 μl 30 μM Taqman probe, 0.4 μl ROX Reference Dye and 6 μl ddH$_2$O, on the 7500 Real-Time PCR System (Applied Biosystems, USA). The conditions were as follows: 95 °C for 10 s, 40 cycles at 95 °C for 5 s, and 60 °C for 34 s.

All samples were analyzed in duplicate, and the values of the sample copies were obtained after quantitative amplification and normalized to GAPDH using $2^{-\Delta CT}$ method, where $\Delta CT = CT(MK) - CT(GAPDH)$ (Livak and Schmittgen 2001). Water was used as negative and quality controls, and each sample was measured in triplicate.

CEA, CA125, and NSE assay

The traditional NSCLC diagnostic markers, CEA, CA125, and NSE, were measured by the chemiluminescent microparticle enzyme immunoassay using the Roche Cobas 6000 analyzer system and commercially available immunoassay kits (Roche Diagnostics GmbH, Germany).

Statistical analysis

Kolmogorov–Smirnov test was performed to determine the distribution of the samples. The results were represented as median (Interquartile range, IQR). Mann–Whitney U test or Kruskal–Wallis test was used to analyze the significance of differences between groups. Receiver operating characteristic (ROC) curves were established to discriminate the subjects with or without NSCLC. The Youden index (sensitivity + specificity −1) was used to determine the optimal cutoff point of circulating MK mRNA levels. The ability of MK mRNA along with CEA, CA125, and NSE to predict the disease outcome of NSCLC versus LEL was verified by multivariate logistic regression analyses, using the "Enter" method with 95 % confidence interval to assess the simultaneous contribution of each covariate in the multivariate analysis. All data were analyzed by SPSS software version 13.0 (SPCC Inc., Chicago, IL, USA), and the significance level was set to $P < 0.05$. All P values were two sided.

Results

MK mRNA level in PBMC from three groups

PBMC MK mRNA level relative to GAPDH mRNA level was evaluated in all the samples. The median values and IQR of MK mRNA level were 3.05 E−03 (IQR: 2.68E−03–4.01E−03) in healthy controls, 3.69E−03 (IQR: 2.89E−03–5.77E−03) in LEL patients, and 7.20E−03 (IQR: 3.82E−03–12.25E−03) in NSCLC patients, respectively. As shown in Fig. 1, upregulated expression of MK mRNA was found in patients with NSCLC compared with patience with LEL ($P < 0.001$) and healthy controls ($P < 0.001$), but there was no significant difference between the latter two groups ($P = 0.073$, Mann–Whitney U test).

Correlation between PBMC MK mRNA and clinical characteristics of NSCLC patients

The relationships between PBMC MK mRNA level and clinicopathological features of patients with NSCLC was demonstrated in Table 1. No association was found between PBMC MK mRNA level and clinical characteristics of NSCLC patients, such as gender, age factors, smoking habit, and pathological types and tumor size. However, higher MK mRNA level was observed in patients with III–IV stage NSCLC than those with I-II stage ($P = 0.026$). In addition, a significantly higher MK mRNA level was observed in patients with poorly differentiated NSCLC than in those with highly or moderately differentiated NSCLC ($P = 0.025$). Higher MK mRNA level was also associated with the lymph node metastasis ($P = 0.022$, Mann–Whitney U test).

Diagnostic value of PBMC MK mRNA level for NSCLC

ROC curve analysis illustrated that PBMC MK mRNA level was a potential biomarker for NSCLC. As shown in Fig. 2,

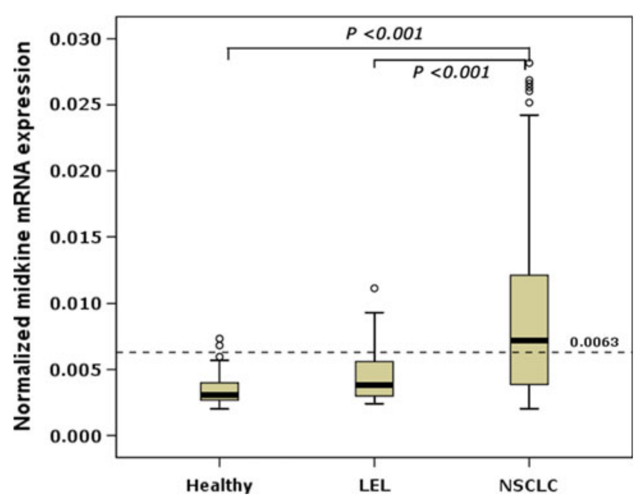

Fig. 1 Box plot for the normalized MK mRNA level in PBMC determined by real-time PCR in all specimens. Significantly elevated MK mRNA level was detected in PBMC from patients with NSCLC compared with those from healthy controls, or patients with LEL (Mann–Whitney U test, $P < 0.001$, respectively)

the AUC values for PBMC MK mRNA in patients with NSCLC versus healthy individuals and patients with LEL were 0.814 (95 %CI = 0.738–0.894) and 0.759 (95 %CI = 0.673–0.846), respectively. Moreover, the optimal cutoff point calculated by Youden's index was 0.0063, providing a sensitivity of 57.47 % and a specificity of 93.33 % between NSCLC and LEL patients, and a sensitivity of 56.32 % and a specificity of 93.33 % between NSCLC patients and healthy volunteers, respectively (Table 2).

To compare the diagnostic value between MK mRNA and the other tumor markers on the diagnosis of NSCLC, the levels of such as CEA, CA125, and NSE were also analyzed. The results indicated that the levels of all tested tumor markers were significantly higher in patients with NSCLC than in patients with LEL and normal subjects (data not shown). The AUC for PBMC MK mRNA, CEA, CA125, and NSE was displayed in Table 3. The AUC for PBMC MK mRNA was significantly larger than that for CA125 or NSE and had the similar diagnostic value with that for CEA to differentiate NSCLC patients from patients with benign lung and healthy individuals. These data indicate that MK mRNA is a high-performing biomarker for NSCLC.

Furthermore, multivariate logistic regression analysis was performed with the presence of NSCLC as the binary dependent variable and the MK mRNA, CEA, CA125, and NSE as covariates (Table 4). Our data suggest that PBMC MK mRNA level above the cutoff value presented a chance of 11-fold higher for NSCLC occurrence.

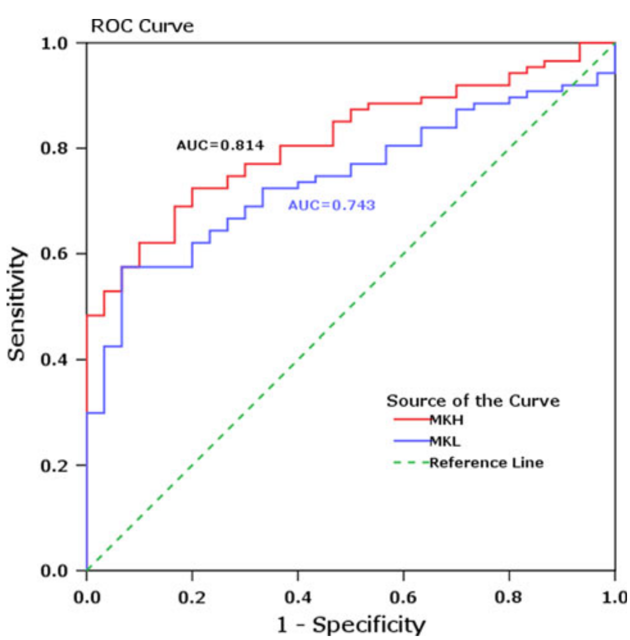

Fig. 2 ROC curve analysis of MK mRNA level for the differentiation of healthy individuals and patients with LEL from patients with NSCLC. The AUC was 0.814 (MKH: patients with NSCLC vs. healthy individuals) and 0.743 (MKL: patients with NSCLC vs. patients with LEL), respectively

Discussion

In this study, we employed highly sensitive and specific real-time PCR to detect PBMC MK mRNA level in NSCLC patients with LEL patients and healthy volunteers as controls. Our data showed that PBMC MK mRNA level was significantly higher in patients with primary NSCLC than in other groups. In addition, we found that higher MK mRNA level was correlated with clinical stages, differentiation, and lymph node metastasis of NSCLC. AUC of MK mRNA could differentiate NSCLC patients from patients with benign lung and healthy individuals. Furthermore, multivariate analysis indicated that PBMC MK mRNA level above the cutoff value presented a chance of 11-fold higher for NSCLC occurrence.

MK was *originally identified* as a gene overexpressed in the mid-gestational period during embryogenesis (Kadomatsu et al. 1988). In the past few years, MK mRNA has been shown to be overexpressed in various human malignant tumors, such as pancreatic (Ohhashi et al. 2009), gastric (Huang et al. 2007), colorectal (Ye et al. 1999), ovarian (Nakanishi et al. 1997), bladder (O'Brien et al. 1996), gastrointestinal carcinomas (Aridome et al. 1995), and neuroblastoma (Fiegel et al. 2008), oral squamous cell carcinoma (Ota et al. 2008), as well as lung cancer (Garver et al. 1993; Zhang et al. 2009). These studies have suggested that MK plays an important role in tumorigenesis, development and metastasis of tumors, and it could serve as a novel tumor marker. Consistent with this idea, in the present study we found that the level of MK mRNA in PBMC from patients with NSCLC was markedly higher than that from patients with LEL and healthy donors, suggesting that PBMC MK mRNA level may be a potential biomarker for monitoring the progress of NSCLC.

We further investigated the relationship between PBMC MK mRNA expression and clinical characteristics of NSCLC. The results showed that higher MK mRNA expression was correlated with clinical stages ($P = 0.026$), differentiation ($P = 0.025$), and lymph node metastasis ($P = 0.022$). However, it was independent of age, gender, tumor size, and smoking habit. Notably, ROC curve analysis indicated that MK mRNA expression was a potential sensitive and specific marker for diagnosis in NSCLC disease. Compared with traditional NSCLC diagnostic markers, CEA, CA125, and NSE, MK mRNA is as good or more appropriate for the detection of NSCLC.

Previous studies have demonstrated a plethora of tumor markers for lung cancer like CA125, CA199, CA153, CEA, SCC, and NSE (Molina et al. 2008). These markers are conventionally used for tumor screening, monitoring, and prognosis, although their diagnostic specificity is not especially high. Li et al. (2012) reported that levels of CEA, CA125, and NSE in patients with pathologically

Table 2 ROC curves for MK mRNA Level in PBMC to distinguish NSCLC patients

Compared to NSCLC patients	Cutoff value	Sensitivity (%)	Specificity (%)	AUC	*P* value*	95 % CI[a]
Healthy volunteers	0.0063	56.32	93.33	0.814	<0.001	0.738–0.891
LEL patients	0.0063	57.47	93.33	0.759	<0.001	0.673–0.846

* Asymptotic significance, null hypothesis: true area = 0.5

[a] 95 % confidence interval of the difference

Table 3 Comparison of AUCs for PBMC MK mRNA, CEA, CA125, and NSE

Marker	NSCLC versus healthy controls			NSCLC versus LEL patients		
	AUC	95 % CI	*P*[a]	AUC	95 % CI[a]	*P*[b]
MK mRNA	0.814	0.738–0.891		0.759	0.673–0.846	
CEA	0.768	0.684–0.851	>0.05	0.736	0.648–0.823	>0.05
CA125	0.649	0.541–0.757	0.014	0.628	0.525–0.731	0.029
NSE	0.647	0.535–0.758	0.016	0.623	0.514–0.732	0.027

[a] 95 % confidence interval of the difference

[b] *P* value for the difference between AUCs of MK mRNA and other markers

Table 4 Results of multivariate analysis

Covariate	OR	SE	95 % CI	*P*
MK mRNA	11.183	0.616	3.345–37.387	0.000
CEA	5.918	0.585	1.881–18.626	0.002
CA125	2.033	0.598	0.629–6.567	0.236
NSE	2.449	0.724	0.592–10.132	0.216

Logistic regression analysis was performed with the presence of NSCLC as binary dependent variable, with the levels of MK mRNA, CEA, CA125, and NSE as covariates

OR odds ratio, *SE* standard error, *CI* confidence interval

confirmed lung cancer were significantly higher than those in patients with benign pulmonary disease or control subjects. Ferrigno et al. (2003) clarified that patients with elevated serum concentration of NSE are a high-risk group of NSCLC patients with shorter survival. They suggested the serum level of NSE in all patients with NSCLC should be determined, at least at the diagnostic level. In our study, to evaluate the potential diagnostic value for NSCLC, conventional tumor markers including CEA, CA125, and NSE were compared with MK mRNA both in patience with NSCLC versus healthy controls and in patients with NSCLC versus patients with LEL. ROC analysis showed that MK mRNA is more useful than the conventional biomarkers CA125, NSE in either comparison. Since no significant correlation was noted between the PBMC MK mRNA and CEA and these two markers were independent predictive factors in multivariate analysis, it is possible that a combination assay with MK mRNA and CEA may be more useful in the improvement of diagnosis of NSCLC. Based on AUC of MK mRNA and predictive capacity of MK mRNA for NSCLC occurrence, we propose that MK mRNA is valuable to differentiate NSCLC patients from patients with benign lung and healthy individuals.

In conclusion, our study demonstrates that the detection of MK mRNA in PBMC by quantitative real-time PCR is a simple and non-invasive approach and may be used for diagnosis of NSCLC. However, further studies enrolling larger samples of NSCLC patients should be performed to confirm our findings, due to the small size of samples in this study. These investigations will provide important information on the potential of MK mRNA as a non-invasive marker for NSCLC patients.

Acknowledgments This work was supported by grant from General Science and Technology Project Fund of Huzhou City (No. 2009YS01). We thank the staff at Department of Thoracic Surgery and Department of Respiratory Medicine in Huzhou Centre Hospital for their assistance.

Conflict of interest We declare that we have no conflict of interest exits in the submission of this revised manuscript entitled “Midkine mRNA level in peripheral blood mononuclear cells is a novel biomarker for primary non-small cell lung cancer: a prospective study,” and the manuscript is approved by all authors for publication. I would like to declare on behalf of my coauthors that the work described was original research that has not been published previously and not under consideration for publication elsewhere, in whole or in part.

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